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Traceable measurements of the static permittivity of dielectric reference liquids over the temperature range 5–50 °C

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Abstract

Measurements of the static permittivity of pure samples of eight polar liquids—acetone, butan-1-ol, dimethyl sulphoxide, ethanediol, ethanol, methanol, propan-1-ol and water—and two non-polar liquids—cyclohexane and silicone oil—were measured in a shielded micrometer-driven, parallel-plate admittance cell at temperatures from 5 to 50 °C. The experimental method, which requires measurements of capacitance using a four-terminal-pair impedance analyser, is described in detail. The full measurement results have been tabulated separately in a National Physical Laboratory report. The present paper provides a comparative study of these latest results with previously published data and also describes the details of the uncertainty analyses employed. Admittance cell measurement techniques are also briefly reviewed to put this work into context.

Keywords: admittance cell measurements, dielectric liquids, dielectric parameters, permittivity measurements, reference materials, static permittivity

1. Introduction

Reference materials are needed for calibrating and checking instruments that are used for dielectric measurements [1]. Liquids can make particularly good reference materials, since they are completely uniform and homogeneous (unlike solid polymers, for example). If they are of high purity they vary negligibly in their dielectric properties between one sample and another. They also have the advantage of being commercially available in suitably pure grades at reasonable cost. Such dielectric reference liquids are commonly used for calibrating coaxial dielectric probes [2, 3]. For health and safety reasons, they are also used to check measurements of the dielectric properties of composite *tissue equivalent* liquids that are employed for determining the specific absorption rate (SAR) of electromagnetic power emitted from mobile telecom devices such as cell phones [4].

For the effective use of reference liquids, there is a general requirement for published information on their dielectric properties based upon measurements that are accurate, traceable and reliable. Unfortunately, much of the existing published data in the scientific literature on the dielectric properties of potential reference liquids fails to meet the required criteria. Uncertainty statements in many papers are not backed by evidence and the traceability of measurements is not made clear. In some instances there are significant discrepancies between the reported results for a given liquid, as demonstrated in table 1 of [5]. The programme of measurements reported on in this paper aimed to overcome these problems and to meet the ongoing need for accurate traceable data on dielectric reference liquids. The measurement results have already been published in a report [6], which contains both 'static' (see below) and RF and microwave (30 MHz–5 GHz) data. This paper describes the experimental techniques used to produce the 'static' permittivity data and the uncertainties that were derived for the measurements—a separate paper on the RF and microwave methods and measurements is under preparation. Some aspects of the experimental techniques have already been reported [7, 8]. The report [6] and the guide [1] both contain

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recommendations on the correct use of dielectric liquids as reference materials.

Note that the term 'static' permittivity is used here to describe the real part of the permittivity measured in the frequency range 0.1–1.0 MHz. (For certain liquids, a subset of this range has been used to minimize uncertainty). Within the precision of measurement, there was no observable change of real permittivity in this range in any of the liquids discussed here. No change would be expected at still lower frequencies so this parameter may therefore be taken as the 'static' permittivity parameter that is used in dielectric relaxation models (such as the Debye relaxation [9]) to describe dielectric properties as a function of frequency (see, e.g., [1]).

The static permittivity is obtained from measurements of the capacitance of a cell in which the liquid samples are placed. If the electrodes are surrounded by a uniform dielectric, the total measured capacitance C_t is related to the real part of the relative permittivity ε' by [1]:

$$C_t(d) = \varepsilon' C_g(d) + \varepsilon' C_f(d) + C_s \tag{1}$$

where C_g is the vacuum 'geometric' capacitance between the electrodes, C_f represents the capacitance due to fringing fields which occur around the edges of the electrodes and C_s represents constant 'stray' capacitances in parallel with the whole electrode system and cell. C_g, by definition, is inversely proportional to the separation of the electrodes and, in the working regime of a dielectric measurement cell, is the dominant term. In the case of a parallel-plate-electrode cell it is given by

$$C_g(d) = \varepsilon_0 \varepsilon' A/d \tag{2}$$

in which A represents the area of the electrodes, d is their separation, ε' is the real part of the permittivity of the dielectric between the electrodes and ε_0 is the permittivity of free space. C_f is a potentially calculable or measurable function of d, but here it may remain an unspecified parameter because the experimental methods considered in this paper do not require C_f to be known. Its magnitude is typically 5–10% of C_g. The stray capacitance C_s can be made negligible by careful shielding around the cell.

Parallel-plate cells with circular electrodes are widely used for measuring the permittivity of disc-shaped solid samples. In such measurements, significant errors can arise from fringing fields, since the sample cannot completely enclose the electrodes (i.e. some of the fringing fields pass through air). The fringing capacitance C_f can be much reduced by the addition of a third electrode that is referred to as a guard ring. The guard ring surrounds the low voltage or 'low' electrode, and is connected to the shield terminal of the measurement instrument. For measurements on liquids, this configuration has the advantage of requiring smaller sample volumes [10]. For planar specimens (which do not fill the annular space between the guard ring and 'low' electrode) corrections are needed for accurate measurements. A popular method for making these corrections is to define an 'effective area', which is a function of the sample permittivity [11, 12]. Three-terminal cells become unsatisfactory at frequencies above a few megahertz (for an electrode diameter of approximately 50 mm) because of inductive effects.

Accurate measurements on liquids, as opposed to solids, may be undertaken without any requirement for a guard ring by using a two-terminal cell with a surrounding metal shield. This is the approach adopted here. If the shield is in the form of a container that is completely filled by the sample, it follows directly from (1) that, if C_s is negligible, the permittivity of the liquid is the ratio of the inter-electrode capacitance of the cell when it is filled by a sample to that when it is filled by a vacuum—assuming that both are measured with the same electrode separation. To be consistent with other publications, the nomenclature 'shielded two-terminal cell' is adopted here to describe this kind of cell.

There are published admittance cell designs using parallel-plate [11, 13–15] and cylindrical geometries [16–19], both of which may be used in two- and three-terminal (electrode) configurations. In [18], an admittance cell based on Thompson and Lampard's 'cross-capacitance' theorem [20] is proposed. This allows the liquid to be contained within a thin-walled cylindrical tube on which the electrodes are deposited externally.

If admittance cells are used for variable-temperature measurements, the effects of expansion must be considered. These may be minimized by constructing the cell from low-expansion-coefficient materials. This is most easily achieved in designs in which the inter-electrode distance is fixed. One approach [16, 18] is to use low-expansion-coefficient glass on which electrodes have been formed by deposition. Note, however, that a *variable* electrode-separation cell has been used in the present work because it helps one to estimate and overcome several systematic measurement errors (see below).

Four-terminal-pair measurement is the best approach for determining inter-electrode capacitance at low frequencies because it is practically unaffected by other capacitances in the system, including those between the measurement electrodes and the shield or guard ring. Historically, AC bridges [20–22] were generally used for such capacitance measurements. They are still used for the highest precision measurements, but for most applications, *LCR* meters and impedance analysers are these days more convenient to use and they have adequate resolution for the measurements described here.

Resistive losses in thin-film electrodes and at the junctions between the electrodes and the leads connecting them to the impedance analyser need to be considered. Corrections for electrode resistance have been analysed [23], but if high conductivity electrodes are used they are often unnecessary [16].

2. Description of the cell

The admittance cell described in this paper (figure 1) is of a shielded parallel-plate (two-terminal) design derived from that devised at NPL in the 1930s by Hartshorn and Ward [13]. A precision large-barrel (100 mm diameter) micrometer is used for adjustment of the gap between the electrodes. The ability to perform measurements over a range of gap sizes helps one to assess and reduce uncertainties, and also enables correction factors for expansion or contraction of the cell to be calculated. The cell was first used in an earlier NPL programme of liquid measurements over 20–30 °C [5]. To allow satisfactory use over an extended temperature range, a number of modifications have been made to it.

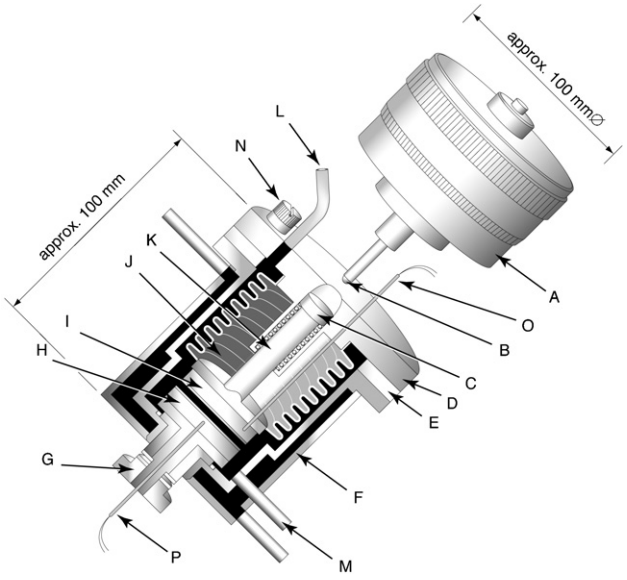


Figure 1. The admittance cell. Key: A: micrometer, B: ball-bearing, C: PEEK thermal insulator, D: upper assembly, E: PEEK electrical and thermal insulator, F: screen/shield with internal water duct for temperature control, G: 'low' electrode terminal, H: 'low' electrode, I: 'high' electrode, J: bellows, K: drive rod for 'high' electrode, L: riser for liquid under test, M: inlet for liquid under test, N: 'high' electrode terminal, O, P: platinum-resistance thermometers (removed during admittance measurements). For clarity the cell is shown in partly 'exploded' form. When in use ball-bearing B contacts the PEEK insulator C. The support arrangement for the micrometer A is also not shown here.

In figure 1, the liquid under test is contained within a metal-shield cell surrounding the disc electrodes. The electrodes (H and I) and the shield (F) are made from gold-flashed brass. The diameter of the electrodes is 40 mm. Bellows (J) are used to allow the top electrode to be moved axially, while maintaining a liquid seal. The large-barrel micrometer (A) allows precise electrode positioning (resolution 0.0005 mm). Water is circulated through a duct within the walls of the shield from a temperature-controlled bath to control the temperature of the liquid being measured. Connections to an impedance analyser (Hewlett-Packard model HP4194A), which has been used for making the measurements of capacitance, are made via matched coaxial cables. The 'potential' and 'current' cables are connected at the electrodes. The cable screens are connected to the shield (F). A metal cover at shield potential (not shown) is placed over the lower low-voltage ('low') electrode connection (G) to reduce stray capacitances outside the cell. An insulator (fused silica) separates this electrode from the shield. The whole of the upper high-voltage ('high') electrode (I), the bellows (J), top plate (D) and micrometer (A) assembly are connected to the 'high' terminal on the impedance analyser. This assembly is insulated from the shield by a layer of PEEK (polyetheretherketone) (E).

A disc (C) made from a low expansion-coefficient polymer (PEEK) between the anvil of the micrometer and the top electrode reduces heat conduction into the barrel of the micrometer. The figure shows the cell in partly 'exploded' form—in practice disc C is forced into contact with the ball-bearing B by a spring (shown around the shaft K) and is driven by the micrometer. The upper assembly (D) of the cell has a weak thermal connection to the remainder of the cell, so to attain equal temperatures at the two electrodes it is necessary to control its temperature actively. This was achieved by clamping a loop of copper tube to it through which temperature-controlled air flowed. Water circulation could not be used in this case since the upper assembly (D) is electrically connected to the 'high' electrode, which must remain electrically isolated. For the temperature control of D, air from a compressed-air laboratory supply was heated or cooled using a vortex tube (a passive device with an input port and two output ports, one giving heated air and one giving cooled).

With these improvements to the cell, it was possible to undertake measurements over the temperature range 5–50 °C. Calibrated platinum-resistance thermometers 1 mm in diameter (O, P) were used for measuring the electrode temperatures. They were always removed from the cell immediately prior to electrical measurements to prevent the residual admittances associated with them from affecting the measurements.

Figure 2 shows the electrical connections between the terminals on the front panel of the impedance analyser, and the cell. Extra shielding around the 'low' terminal ensures that there are no stray fields around the outside of cell. The outer and inner conductors of the coaxial cables were separated as close to the terminals of the cell as possible, and during an experiment they were always left undisturbed. The measured total inductance of these cable links, L, was approximately 42 nH. Apparent measured capacitance C_{app} is related to the true capacitance C in the cell at frequency f by the approximate formula:

$$C_{\mathrm{app}} = \frac{C}{1 - 4\pi^2 f^2 LC}. \tag{3}$$

The largest capacitances encountered in this programme of measurements (for water) were approximately 1500 pF. This series LC combination will produce a change of $(0.2 \pm 0.1)\%$ in the measured capacitance at 1 MHz. The error drops rapidly with reducing frequency and is only $(0.06 \pm 0.02)\%$ at 500 kHz. Measuring smaller capacitances also reduces the percentage change. Data are easily corrected to obtain true capacitance. Uncertainties can be minimized by suitable choice of maximum operating frequency. In the work presented, lead-inductance effects were only a significant issue for measurements on water as all of the other measured liquids had lower permittivity.

At low frequencies electrode-polarization effects [24] can cause large measurement errors in liquid dielectric measurements but they can become negligible above typically 100 kHz (depending on the liquid). Table 1 shows measurements of capacitance of a sample of methanol in the admittance cell as a function of frequency.

Electrode polarization has greatest effect at the smallest electrode separations. Platinum-black electrodes, which on a microscopic scale have increased surface area compared to plain metals, are sometimes used to reduce these effects. Various correction formulae have been proposed [25–27]. The approach used in this work is to employ gold-flashed-brass electrodes and to avoid these errors by measuring at higher frequencies (above 100 kHz but below 1 MHz on account of inductive effects).

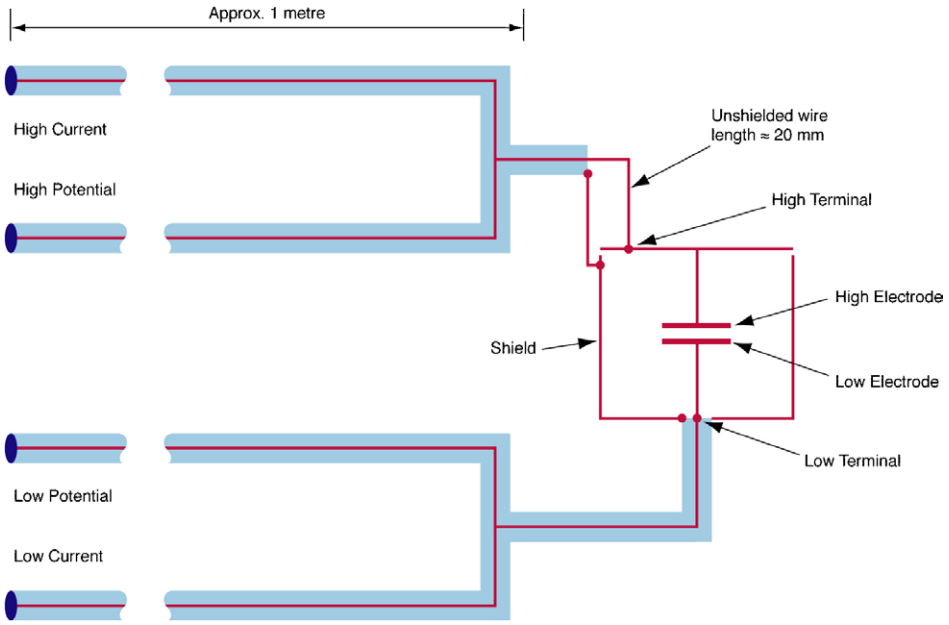


Figure 2. Electrical connections between the impedance analyser terminals (at the left) and the cell (on the right). The lines with grey background are coaxial lines, the outer conductors of which were connected to the cell shield, as shown.

Table 1. Measured apparent capacitance of a methanol sample over a wide frequency band at fixed electrode separation. At frequencies below 100 kHz the electrode-polarization effect increases the capacitance significantly. At 10 MHz lead inductance causes the apparent capacitance to rise. These two effects are insignificant only in the range 100 kHz–1 MHz.

Frequency	C (pF)
100 Hz	15 460.0
200 Hz	4 540.0
500 Hz	1 145.0
1 kHz	572.5
10 kHz	363.8
100 kHz	360.20
200 kHz	360.28
500 kHz	360.07
1 MHz	360.06
10 MHz	360.30

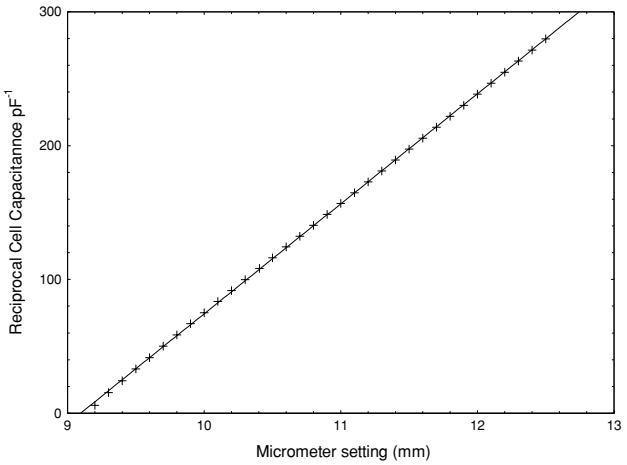


Figure 3. The micrometer reading M plotted against the reciprocal of the cell capacitance for the air-filled cell. The intercept on the horizontal axis gives the 'effective' micrometer zero-error, Z, as 9.095 mm (see the text). Note the departure from linearity of measured data at the left (i.e. for small electrode gaps).

3. Experimental technique

As stated above, for a fully shielded cell the static permittivity is, by definition, the ratio of capacitances measured with and without the sample (i.e. with the cell vacuum-filled in the latter case) *assuming that the dimensions of the cell remain unchanged between the two measurements*. Now the dimensions of the cell change with temperature, but the cell design does not permit sufficiently effective control of the temperature of the top electrode unless a liquid sample is present, so measurements of the empty cell (i.e. on air) were always made when the cell was uniformly at room temperature. A practical problem, therefore, arose over how to perform the measurements at any other temperature than room temperature. This was overcome by means of an extrapolation technique (described below) that enables corrections to be made for the change in cell dimensions between measurements on air and on liquids at other temperatures.

In these circumstances, an important parameter to determine is the *effective* micrometer zero-error Z, given by $M = d + Z$, where M is the actual micrometer reading and d is the electrode separation. Z normally has an arbitrary value depending on how the cell has been assembled, but in any practical cell it is a function of temperature. Note that Z is unlikely to be the same as the *mechanical* zero error, defined as the micrometer reading at which the electrodes just touch, because of the roughness and non-parallelism of the electrodes. The 'effective' zero error of the micrometer is that which corresponds to an extrapolated *infinite* total cell capacitance, i.e. to *zero* reciprocal capacitance, see figure 3.

Linear extrapolation of the reciprocal of capacitance measurements made at a number of typical micrometer readings, M_i, enables the micrometer reading that corresponds to Z to be found. By applying this procedure to both empty cell and sample measurements, changes of the electrode separation in the cell due to expansion or contraction with temperature can be determined. Points at small electrode spacing (i.e. $d < 0.75$ mm) are excluded from the calculation, partly because these were never actually used in the permittivity measurements but more importantly, because in this region the curve becomes nonlinear as a consequence of the electrodes not being perfectly parallel. The fringing capacitance term $C_f(d)$ (see equation (1)) will not be an exact reciprocal function of d and is another cause of nonlinearity.

In permittivity measurements, this electrical approach to measuring micrometer zero error is better than any mechanical method because it determines static permittivity via the very parameter that matters most in these permittivity measurements, namely the cell capacitance. Note that the gradient of the curve in figure 3 should be proportional to the permittivity of the material filling the cell (air or sample); however this approach to measuring permittivity based upon the ratio of these gradients is not followed here. To gain additional information about the measurement uncertainties, and to reduce the dependence of results on micrometer linearity, an alternative procedure and analysis is used.

Measurements on the liquids were made only when the two thermometers at the electrodes both indicated the expected temperature to within 0.02 °C of the required temperature. This was achieved by careful adjustment of the controls of the water bath, and the input air pressure to the vortex tube. Lagging was used to improve the temperature uniformity. Typical micrometer positions correspond to a range of electrode gaps, d_i, from 2.75 to 0.75 mm at 0.25 mm intervals. The sequence of the measurements was always the same, with the largest gap being measured first. Measurements at nine micrometer positions took a few minutes (the maximum practical number of measurements being limited by temperature drift). As stated above, the thermometers were removed during each measurement, but they were re-inserted afterwards, and the final temperature recorded.

Under computer control, the data were extracted from the impedance analyser and stored for processing. The instrument averaging was set to 128, and the integration time to 'medium'. The processing algorithm is shown in figure 4. Input quantities are the reciprocals of the measured capacitances (corrected for lead inductances and stray capacitance), R, at each frequency, the corresponding micrometer positions, M, and the weighting factors, W, for both micrometer positions and frequency (details are given in the caption to figure 4).

The process described above for determining the 'effective' micrometer zero error for air and sample measurements uses a weighted least-squares fit of reciprocal capacitance as a function of micrometer position (stage 1 in figure 4). The residual errors of the fit are checked in case there are any erroneous points in the data. Measurement points that are found to fall in a nonlinear region are excluded by giving them zero weighting. This calculation is performed independently at each frequency; then the intercepts are

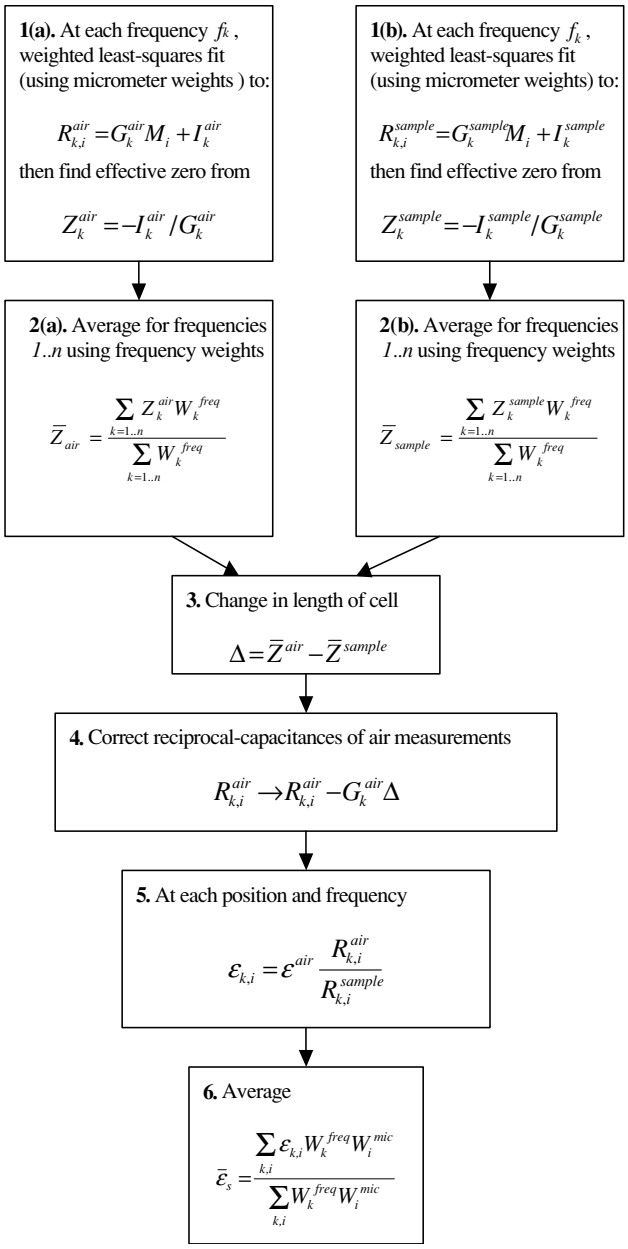


Figure 4. Calculation procedure from reciprocal-capacitance data ($R_{k,i}$) and micrometer positions (M_i). Index k is used for frequencies and i is used for micrometer positions. The permittivity of air (ε^{air}) is unity for practical purposes (its value is approximately 1.0006). Other symbols are G (the curve gradient, see figure 3), I (its intercept), W^{mic} (micrometer position weighting factor), W^{freq} (frequency weighing factor), Z the 'effective' micrometer zero error. The weighting factors are discussed in the text.

averaged (stage 2) to obtain an average 'effective' zero error. The standard deviation of the intercepts calculated at each frequency is typically ±0.0005 mm.

The change in the length of the cell between the air and liquid measurements is then calculated (stage 3). The measured reciprocal capacitances for the air measurement are then corrected to the values that they would have if the cell had the same length for both air and liquid measurements (stage 4). A value for liquid permittivity is then found from the capacitance ratio at each micrometer setting for each frequency (stage 5). Note that this ratio should be multiplied by the

permittivity of air (typically 1.0006) to obtain the correct liquid permittivity. Since the actual permittivity of the liquids concerned does not alter significantly over this frequency range, all the results are then combined in a weighted average (stage 6).

The distribution of the results produced at stage 5 can be inspected to reveal systematic trends in the impedance measurements, such as an increase in measured capacitance at the lowest frequency due to electrode polarization. If there is evidence of such effects, the weightings can be adjusted to take account of them, and the data re-processed.

Since the reference (air) and sample measurements are carried out at different temperatures, a correction for the change in area of the electrodes due to expansion is required. If T_1 and T_2 are the temperatures of the air and sample measurements, sample capacitance data must be divided by $(1 + \alpha(T_2 - T_1))^2$. The linear expansion coefficient α of the electrodes (brass) is taken to be 1.75×10^{-5} K^{-1} [28]. For water, the highest permittivity liquid, the resulting correction to permittivity is –0.07 at 50 °C, so this always amounts to a small correction in this set of measurements.

To check for problems such as temperature drift and the presence of air bubbles, the first and last measurements in each measurement session on a liquid sample were made at 20 °C and the resulting computed permittivities compared. All samples were taken from newly opened bottles of high-purity grades which were obtained from chemical suppliers (further information is given in [6]).

4. Results

4.1. Evaluating the impedance analyser and the experimental method

The two main requirements relating to the performance of the impedance analyser are that it should be adequately stable and should be able to measure capacitance ratios of the order of ε' sufficiently accurately. Absolute accuracy in measuring capacitance is not necessary because ε'—the *relative* permittivity of the liquid—is a dimensionless quantity, which is here measured as a ratio. Preliminary experiments were therefore undertaken to assess the stability and linearity of the analyser: both are taken to be sources of uncertainties in the measurement—see section 5. One way in which the linearity has been tested is to measure standard capacitors on the analyser that have been calibrated traceably to international standards on a bridge traceable to international standards. These measurements were performed over the range 1 kHz–1 MHz, which is the upper limit for these large-dimension capacitors. In order to test linearity across the complete measurement frequency band, a number of small Sufflex capacitors were subsequently measured across the frequency range. They had capacitance values corresponding to typical cell capacitances when the cell is both empty and when it is filled with liquid. The point of using small capacitors (∼5–8 mm linear dimensions) on short leads for this exercise is that their inductance is small and so their measured capacitances should change only very slowly with frequency, and in fact should change negligibly over the band of interest. These tests demonstrated that the analyser was adequately linear for this programme of measurements and also complied with the manufacturer's specification. For safety it was decided to take this specification as the uncertainty for the linearity, which results in an uncertainty contribution of $\pm 0.0003 \times \varepsilon'$ in ε', as discussed further in section 6.

To gain further confidence in the capabilities of the impedance analyser, measurements were also performed with modifications as follows: (1) with a 1200 pF capacitor soldered between the 'high' electrode and the screen, and (2) with an additional 50 cm length of coaxial cable inserted in the 'high' electrode current connection. Neither experiment produced any significant change in important measurement results, e.g. no significant changes in the gradient or intercepts of figure 3 were observed.

Measurements of the capacitance of the empty cell were made in the frequency range 100 kHz–1 MHz and plotted against micrometer position to assist in evaluating the performance of the impedance-analyser/cell combination. Dependence of the capacitance on frequency at each micrometer setting can be used to indicate inductive effects, or measurement errors in the impedance analyser. This can be tested by a plot of residual errors (i.e. measured capacitance − average capacitance), as shown in figure 5. The figure does not show any significant dependence on frequency, although there is some unexpected structure (indicated by arrows), which has been attributed to digitization in the impedance analyser. The spread of the residuals is very approximately proportional to the inter-electrode capacitance.

4.2. The liquids

All samples were taken from newly opened bottles of high-purity grade liquids which were obtained from chemical suppliers. The liquids were chosen for a number of reasons–partly to give a spread of permittivities and (at RF & microwave frequencies) of loss. Ethanol and methanol are commonly used as reference liquids and so it was important to include them, while propan-1-ol and n-butanol offer somewhat lower permittivities. Acetone is a widely used, low toxicity liquid with low loss. Ethanediol has a similar relaxation frequency to ethanol (see the companion paper on RF & microwave measurements upon these liquids) but has a higher permittivity. Dimethyl sulphoxide was chosen because of its match in permittivity to a number of tissue equivalent materials and its relatively low toxicity. However, anyone using this liquid should be aware that exposure to the atmosphere causes the observed permittivity to increase significantly in quite a short time as it is very hygroscopic. Unlike the organic liquids measured here, de-ionized water has already been adequately and widely characterized elsewhere (see below). Furthermore, its permittivity (approximately 80 at 20 °C) is outside the permittivity range for which the cell was designed (4–50). However, it was included to add confidence to the measurements that have been performed in the cell. Cyclohexane is a low-permittivity standard reference liquid (less toxic than benzene which was widely used for many years) as is 1 cS (i.e. centiStokes) silicone fluid which has been widely used in the dielectric metrology of polyethylene because of its close match in permittivity.

Further information on the liquids and their use as reference materials is given in [6].

Table 2. A complete set of methanol measurements at 20 °C. The final result ($\varepsilon' = 33.62$ for this particular experiment) is a weighted average of 200 kHz–750 kHz data at micrometer readings from 7.75 mm upwards. The 'effective' zero error, Z, is 6.82 mm which may be subtracted from the micrometer reading to give the inter-electrode separation. The same data are plotted in figure 6.

Micrometer	Measured permittivity						
M (mm)	100 kHz	150 kHz	200 kHz	250 kHz	500 kHz	750 kHz	1 MHz
7.25	33.895	33.841	33.809	33.740	33.734	33.729	33.742
7.50	33.680	33.641	33.623	33.579	33.577	33.580	33.590
7.75	33.711	33.684	33.667	33.634	33.641	33.639	33.655
8.00	33.649	33.626	33.621	33.593	33.598	33.608	33.619
8.25	33.673	33.646	33.626	33.614	33.618	33.628	33.649
8.50	33.682	33.663	33.637	33.614	33.629	33.636	33.657
8.75	33.681	33.657	33.636	33.616	33.627	33.633	33.650
9.00	33.668	33.646	33.623	33.604	33.612	33.619	33.638
9.25	33.682	33.655	33.643	33.616	33.623	33.629	33.648
9.50	33.682	33.659	33.644	33.617	33.623	33.627	33.640

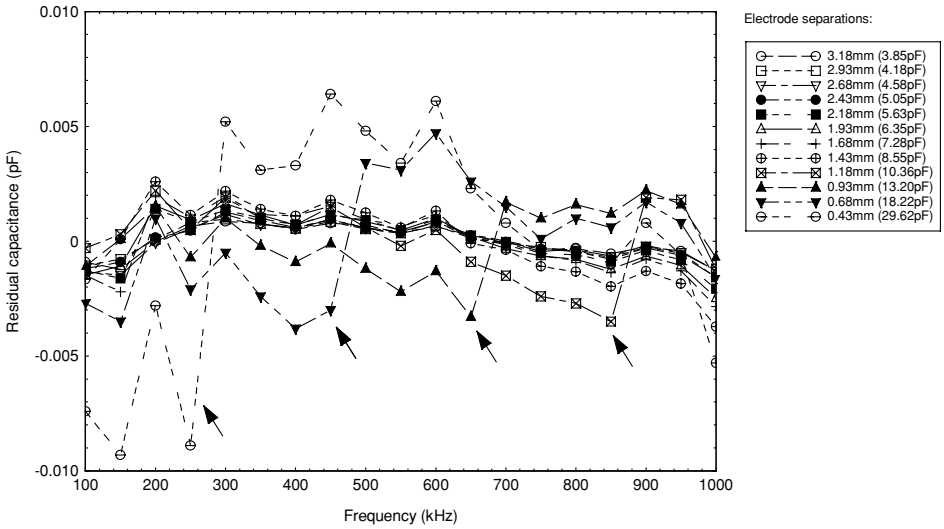


Figure 5. Residual capacitance (i.e. measured − average) data plotted for a number of gap sizes as a function of frequency for air-filled cell. The features indicated on the graph are referred to in the text.

5. Static permittivity results

Typical permittivity results for methanol at stage 5 of figure 4 are shown in figure 6 and tabulated in table 2 as a function of electrode separation and frequency. The final result ($\varepsilon' = 33.62$ for this particular experiment) is a weighted average (stage 6, figure 4). Data at the smallest separations (below 0.75 mm) are excluded, since uncertainties in the zero corrections strongly affect the results. Data at frequencies below 200 kHz are also excluded as an increased permittivity is shown, which is attributed to electrode-polarization effects. This effect can be seen to be greatest at small electrode separations.

The results are compared with literature values in tables 3–6. Note that the latest NPL results are quoted at coverage factor $k = 2$ corresponding to a confidence level of approximately 95%. The uncertainty shown in brackets includes an additional contribution for the temperature measurement uncertainty (±0.1 °C in the range 10–35 °C and ± 0.2 °C at 5 and 40–50 °C). In table 3, the complete methanol results are given and compared with literature values where available. It can be seen that good agreement is obtained. The data in [30] were obtained from measurements at 10 MHz using a parallel-plate cell connected directly

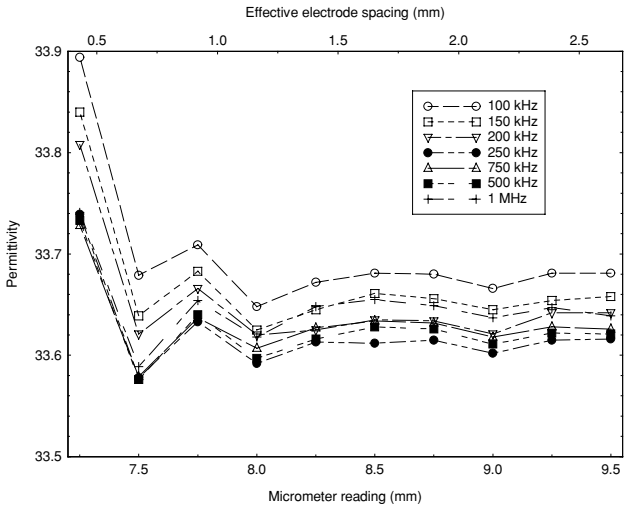


Figure 6. Measured results for methanol at 20 °C as a function of electrode separation for a number of measurement frequencies. The 'effective' zero error, Z, is 6.82 mm which may be subtracted from the micrometer reading to give the inter-electrode separation.

across the terminals of an impedance bridge. The remainder of the literature values were obtained from admittance cell

Table 3. Comparison of the measured static permittivity of methanol with literature values. For notes (a), (b) and (c) on the uncertainties see the caption to table 8.

	NPL new data		Published			
Temperature (°C)	ε'	± [a]	ε	±	Reference	
5	36.86	0.06 (0.07)				
10	35.74	0.06 (0.06)	35.70	0.03 [c]	Kienitz and Marsh	[29]
15	34.68	0.06 (0.06)				
20	33.64	0.06 (0.06)	34.8	0.5 [b]	Jordan *et al*	[30]
			33.66	0.03 [c]	Kienitz and Marsh	[29]
			33.65	0.1 [b]	NPL 1990	[31]
25	32.66	0.06 (0.06)	32.62	0.03 [c]	Cunningham	[32]
			32.66	0.02 [c]	Kienitz and Marsh	[29]
			32.65	0.1 [b]	NPL 1990	[31]
30	31.69	0.06 (0.06)	31.70	0.1 [b]	NPL 1990	[31]
35	30.78	0.06 (0.06)				
40	29.85	0.06 (0.07)	29.86	0.05 [c]	Kienitz and Marsh	[29]
45	28.95	0.06 (0.07)				
50	28.19	0.06 (0.07)				

Table 4. Comparison of the measured static permittivity of cyclohexane with literature values. For notes (a), (b) and (c) on the uncertainties see the caption to table 8.

Temperature	NPL new data		Kienitz and Marsh [29]	
(°C)	ε'	± [a]	ε'	± [c]
10	2.040	0.002 (0.002)	2.039	0.0015
20	2.024	0.002 (0.002)	2.024	0.001
30	2.008	0.002 (0.002)	2.008	0.001
40	1.992	0.002 (0.002)	1.992	0.001
50	1.975	0.002 (0.002)	1.976	0.001

Table 5. Comparison of the measured static permittivity of deionized water with literature values. The admittance cell used in this paper is not of optimum design for measuring water, but these measurements are useful as a check on its operation. For notes (a), (b) and (c) on the uncertainties see the caption to table 8.

Temperature	NPL new data		Kaatze *et al* [33]		Kienitz and Marsh [29]	
(°C)	ε'	± [a]	ε'	± [c]	ε'	± [c]
10	84.04	0.20 (0.20)	83.92	0.20	83.91	0.07
20	80.29	0.20 (0.20)	80.21	0.20	80.16	0.05
30	76.68	0.20 (0.20)	76.56	0.20	76.57	0.05
40	73.31	0.20 (0.21)	73.18	0.20	73.16	0.04
50	69.99	0.20 (0.21)	69.89	0.20	69.90	0.04

Table 6. New results for the static permittivity of other liquids at 20 °C. For note (a) on the uncertainties see the caption to table 8.

	ε'	± [a]
Acetone	21.13	0.04 (0.04)
Butan-1-ol	18.19	0.04 (0.04)
Dimethyl sulphoxide	47.13	0.06 (0.06)
Ethanediol	41.89	0.06 (0.06)
Ethanol	25.16	0.04 (0.04)
Propan-1-ol	21.15	0.04 (0.04)
Silicone oil (1 cS)	2.307	0.002 (0.002)

measurements. NPL data from an earlier programme of work [31] using the same admittance cell prior to modification [5] are consistent with the new data. The data from [32] were obtained using a cylindrical-geometry cell made from Pyrex glass with deposited electrodes.

Table 7. New results for the static permittivity of other liquids at 25 °C. For note (a) on the uncertainties see the caption to table 8.

	ε'	± [a]
Acetone	20.59	0.04 (0.04)
Butan-1-ol	17.53	0.04 (0.04)
Dimethyl sulphoxide	46.49	0.06 (0.06)
Ethanediol	40.75	0.06 (0.06)
Ethanol	24.43	0.04 (0.04)
Propan-1-ol	20.42	0.04 (0.04)
Silicone oil (1 cS)	2.294	0.002 (0.002)

Table 8. New results for the static permittivity of other liquids at 30 °C. Key: (a) NPL data are given at coverage factor $k = 2$, approximately equivalent to 95% confidence level. The figures given in brackets include the temperature contribution. (b) Quoted at 95% confidence level. (c) Assumed to correspond to one standard deviation.

	ε'	± [a]
Acetone	20.10	0.04 (0.04)
Butan-1-ol	16.89	0.04 (0.04)
Dimethyl sulphoxide	45.86	0.06 (0.06)
Ethanediol	39.66	0.06 (0.06)
Ethanol	23.64	0.04 (0.04)
Propan-1-ol	19.75	0.04 (0.04)
Silicone oil (1 cS)	2.279	0.002 (0.002)

As stated above, the admittance cell was designed for measurement of liquids having permittivity in the range 5–40. Both cyclohexane and water, which lie outside this range, were measured to provide further confidence in measurements. The measurements are seen to be in good agreement with IUPAC data (tables 4 and 5) and this certainly adds confidence in the use of this cell for reference liquid permittivity measurements. Summaries of results for the remaining liquids at 20, 25 and 30 °C are given in tables 6–8. Measured results at all measurement temperatures are given in [6].

6. Uncertainties

Uncertainties were for the most part estimated on the basis of the UKAS document M3003 [34], which complies with the

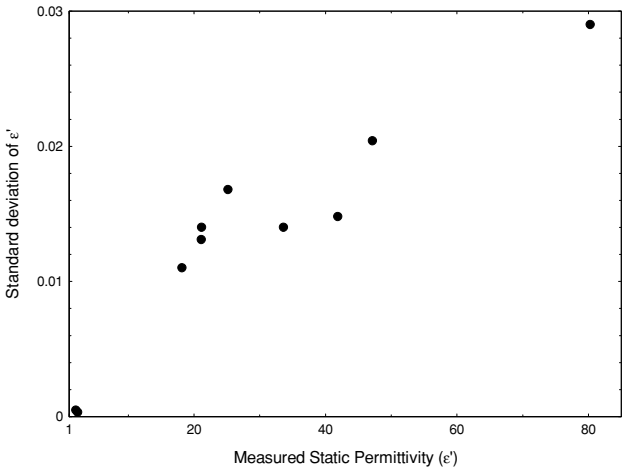


Figure 7. Each liquid was measured in three separate experiments to cover the entire temperature range. Each experiment began and ended with a measurement at 20 °C. Hence, at this temperature there is a total of six final results (calculated in stage 6, figure 4). The above plot shows the standard deviation of the six results as a function of permittivity for the ten liquids.

ISO guide to the expression of uncertainties in measurement [35], commonly called the 'GUM'. However in a number of cases, where uncertainty estimation was difficult, it was decided to err on the conservative side and use worst-case estimates in the uncertainty budget. For almost all of the liquids three separate experiments were carried out, each of which included initial and final measurements at 20 °C. Hence, there are six weighted-average results at this temperature. Figure 7 shows a plot of the standard deviation of these results as a function of permittivity. This graph, which is derived from approximately 60 results, is used as the basis of estimates of the type A uncertainty contributions. This approach provides a more secure basis for estimating type-A uncertainties than the individual sets of data such as that shown in figure 6. In the case of methanol, for example, the type-A standard uncertainty is estimated to be ±0.02. This can be seen to be consistent with the spread of the 200 kHz –1 MHz results for micrometer readings of 7.75 mm and larger in figure 6.

The type-A standard uncertainty of the effective electrode separation in the cell can be calculated (stages 1 and 2, figure 4), and is found to be typically a few micrometres. If correlations are ignored, quadrature addition enables the standard uncertainty in the correction, Δ (stage 3), to be estimated. If there are correlated errors between the lengths of the cell with and without the sample they will to some extent cancel at stage 3 (in the subtraction), so the uncertainty in Δ will be overestimated. The corresponding standard uncertainty in the permittivity is estimated by using a first-order numerical approach. It could be argued that this quantity is already represented by the standard uncertainty derived from figure 7. However, since Δ is derived from measurements under different circumstances (both without and with the liquid filling the cell), it was decided to treat it as a separate contribution, in case there is a correlation with the permittivity of the fluid in the cell (air or liquid sample).

As stated in section 4.1, the absolute accuracy of the capacitance measurements is not significant provided that ratios of capacitance are accurately measured. As explained there, the allowance for impedance-analyser nonlinearity was taken to be $\pm 0.0003 \times \varepsilon'$. Dielectric measurements were only taken when the thermometer readings were stable and within ± 0.02 °C of that specified. However, an allowance for drift has to be made. On the basis of calibration data for the thermometers the total temperature uncertainty contribution was taken to be ±0.1 °C (at coverage factor $k = 2$), except at the extremes of temperature where it is increased to ±0.2 °C since the temperature was more difficult to control. An equivalent permittivity uncertainty is calculated on the basis of the measured temperature dependence of permittivity. The uncertainties for a methanol measurement are summarized in table 9. Similar budgets have been estimated for all the liquids measured.

Table 9. Estimated dominant standard uncertainty contributions for methanol.

Contribution	Standard uncertainty of input quantity	Standard uncertainty of permittivity
Type-A from figure 7		±0.02
Zero-error correction	Approximately ±0.0005 mm	±0.01
Impedance analyser linearity	±0.0003 ε'	±0.01
Temperature[a]	±0.05 °C	±0.01
Temperature[b]	±0.1 °C	±0.02

[a] For 10–35 °C,
[b] For 5 and 40–50 °C

A number of other sources of uncertainty have been considered in these measurements but in comparison to those given in table 9, their effect is generally found to be negligible. They are tabulated in table 10.

7. Discussion

Comprehensive traceable static permittivity measurements for a variety of liquids have been obtained and reported both here and in [6]. Details of the measurements, uncertainty corrections and analytical techniques have been described here in more detail than is conventional in scientific papers in order to demonstrate the clear traceability route and the significant steps taken to reduce measurement errors. It is hoped that these data will prove useful to dielectric metrologists—they are already being used at NPL, e.g. for calibrating coaxial probes and checking other measurements. For a number of the liquids measured the new data are a significant improvement on the data already available. Where suitably robust data were already available in the literature, the results have been compared with data from the extant literature and found to be in good agreement—a fact which boosts confidence in both sets of data.

A major requirement for effective long-term and widespread use of dielectric reference materials is confidence in the values of permittivity attributed to them. Here such confidence has been boosted in a number of ways:

(i) by measuring liquids whose properties are already known sufficiently accurately, namely deionized water and

Table 10. Sources of uncertainty that are estimated in most cases to be negligible in magnitude.

Uncertainty contribution	Comment
Uncertainty in electrode expansion correction	Dependent on linear expansion coefficient. Since the grade of brass is unspecified the standard uncertainty is taken to be 10% of the permittivity correction for expansion. For water at 50 °C the standard uncertainty is estimated to be ±0.007. This contribution is smaller for measurements at lower temperatures and on other liquids.
Contamination of liquid during experiment	Considered to be taken into account in the spread of figure 7, since this contains data from initial and final measurements in experiments which lasted a whole day.
Cable inductance	Standard uncertainty less than 0.001 for methanol.
Electrode polarization	A low-frequency effect which shows strong frequency dependence and increases for small electrode gaps. Data below a chosen frequency (typically 150 kHz) is excluded in the calculation of the averaged result. Any residual effect is taken to be represented in the type A uncertainty
Stray capacitance	Careful shielding used to minimize fields around the outside of the cell. If there were a stray-capacitance effect, it would be most significant at largest gaps, but there is no evidence of this, e.g. in figure 3. It is estimated that $C_s < 0.003$ pF
Change in ε' from DC	For the worst case, butan-1-ol at 10 °C (which has a Debye relaxation frequency of only 170 MHz), the change in ε' is approx. –0.0004

cyclohexane, and showing that the techniques employed here give results that agree well with the literature,

(ii) by using a cell that allows measurements at a number of frequencies and electrode separations, allowing checks to be made for systematic errors and

(iii) by estimating, evaluating and reducing uncertainties (wherever possible) and presenting them in this paper.

An important issue, which cannot be readily answered on the basis of existing literature, is the type of measurement cell that should be used to produce the lowest measurement uncertainties, particularly, for polar (high permittivity) liquids. Although we cannot give a full answer to this question here, we can give an outline of what we consider to be advantages and disadvantages of the cell that we have used. It has the disadvantage of not being fully temperature compensated, so corrections to the cell length must be obtained experimentally. Besides making the measurement procedure more complicated, this requirement gives rise to increased uncertainties. However, the cell design has the advantage that it allows measurements to be made at different inter-electrode separations, which provides more data for assessment of uncertainties and increases confidence in the results.

It is thought that the data presented here should be sufficiently accurate for most uses of dielectric reference liquids; however, improvements are potentially possible, if needed. The spread of results in figure 6 is partly attributable to the behaviour of the impedance analyser. A more precise instrument would help reduce these uncertainties. A number of publications in the literature claim measurement of static permittivity with very low measurement uncertainties, but it is difficult to assess the value of the work, as measurements on recognized reference liquids are not given. We would recommend that some reference liquid data should always be presented in such papers to help advance the experimental techniques, and improve the available data on these liquids.

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